TERM MOCK EXAM

Answers

Check yourself after you finish. Part 2 shows the model answer.

PART 1 · MULTIPLE CHOICE

Q1 **A**	Q2 **B**	Q3 **C**	Q4 **D**	Q5 **D**
Q6 **B**	Q7 **A**	Q8 **C**	Q9 **C**	Q10 **B**
Q11 **A**	Q12 **D**	Q13 **C**	Q14 **A**	Q15 **D**
Q16 **B**	Q17 **B**	Q18 **C**	Q19 **D**	Q20 **A**
Q21 **C**	Q22 **B**	Q23 **D**	Q24 **A**	Q25 **D**

PART 2 · MODEL ANSWERS

26 a) 1 Evaporation · 2 Condensation · 3 Precipitation · 4 Runoff · 5 Percolation (in the order of the numbers on the diagram). b) Transpiration (2 marks). Water vapor rises, cools and condenses into droplets around dust particles (3 marks).

27 a) Grasshopper 20,000 × 10% = 2,000 kJ · mouse 2,000 × 10% = 200 kJ · snake 200 × 10% = 20 kJ (2 marks each). b) Most of the energy at each level is used for cellular processes or released to the environment as heat, so only about 10% is stored in new tissue that the next level can eat.

28 a) Range of tolerance 9–25°C; optimum range 13–21°C. b) Zone of physiological stress: the trout survives but may not grow or reproduce well, and there are fewer fish. c) Abiotic (nonliving).

29 a) Biological magnification: the increasing concentration of toxic substances in organisms as trophic levels increase. b) The toxin is not broken down or excreted; it accumulates in tissues, and each predator eats many organisms from the level below, so the concentration rises at every level and is highest at the top. c) Thin, fragile eggshells, so developing birds die.

30 a) Independent: amount of fertilizer · dependent: plant height. b) Any two: type of plant, amount of water, light, soil, temperature, pot size. c) Bean plants grown the same way with no fertilizer. d) Revise the hypothesis or refine the procedure, then test again.